EXPERIMENT-08

Aim: To develop an Android application that uses Alert Dialog, Date Picker, and Time Picker and handles user interaction.

Theory:
Android provides UI components like AlertDialog, DatePickerDialog, and TimePickerDialog to interact with users. These components help in taking decisions (dialog), selecting dates, and selecting time through user-friendly interfaces.
Android provides interactive UI components like AlertDialog, DatePickerDialog, and TimePickerDialog to enhance user interaction.

- AlertDialog is used to display messages and get user confirmation through buttons like Yes or No.
- DatePickerDialog allows users to select a date using a calendar interface.
- TimePickerDialog enables users to choose time (hour and minute) using a clock interface.

These components help in collecting user input in a simple, structured, and user-friendly way.

1. Design Layout
   - Open activity_main.xml
   - Add:
     - Button → Show Dialog
     - Button → Select Date
     - Button → Select Time
     - TextView → Display result
2. Implement MainActivity
   - Initialize views using findViewById():
     - Buttons and TextView
3. Alert Dialog Implementation
   - Use AlertDialog.Builder
   - Set:
     - Title → "Confirmation"
     - Message → "Do you want to continue?"
   - Add buttons:

    - Yes → show Toast
    - No → show Toast
- Call show()

4. Date Picker Implementation
    - Use Calendar to get current date
    - Create DatePickerDialog
    - On selection:

        - Get day, month, year
    - Display in TextView
5. Time Picker Implementation
    - Use Calendar to get current time
    - Create TimePickerDialog
    - On selection:
        - Get hour and minute
    - Display in TextView
6. Run Application
    - Click buttons:
        - Dialog → shows confirmation
    - Date → opens calendar
    - Time → opens clock

Code:

activity_main.xml

activity_main.xml × MainActivity.java

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        android:gravity="center"
        android:padding="20dp">
        <Button
            android:id="@+id/btnDialog"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Show Dialog"/>
        <Button
            android:id="@+id/btnDate"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Select Date"
            android:layout_marginTop="20dp"/>
        <Button
            android:id="@+id/btnTime"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Select Time"
            android:layout_marginTop="20dp"/>
        <TextView
            android:id="@+id/txtResult"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Result will appear here"
            android:textSize="18sp"
            android:layout_marginTop="30dp"/>
    </LinearLayout>

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

activity_main.xml MainActivity.java

```java
package com.example.experiment_08;

import android.os.Bundle;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.app.AlertDialog;
import android.app.DatePickerDialog;
import android.app.TimePickerDialog;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;
import android.widget.Toast;
import java.util.Calendar;

public class MainActivity extends AppCompatActivity {
    2 usages
    Button btnDialog, btnDate, btnTime;
    3 usages
    TextView txtResult;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        btnDialog = findViewById(R.id.btnDialog);
        btnDate = findViewById(R.id.btnDate);
        btnTime = findViewById(R.id.btnTime);
        txtResult = findViewById(R.id.txtResult);
```

```java
// Alert Dialog
        btnDialog.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                AlertDialog.Builder builder = new AlertDialog.Builder( context: MainActivity.this);
                builder.setTitle("Confirmation");
                builder.setMessage("Do you want to continue?");

                builder.setPositiveButton( text: "Yes", ( DialogInterface dialog, int which) -> {
                    Toast.makeText( context: MainActivity.this, text: "You clicked YES",Toast.LENGTH_SHORT).show();
                });

                builder.setNegativeButton( text: "No", ( DialogInterface dialog, int which) -> {
                    Toast.makeText( context: MainActivity.this, text: "You clicked NO",Toast.LENGTH_SHORT).show();
                });

                builder.show();
            }
        });

        // Date Picker
        btnDate.setOnClickListener( View view -> {

            Calendar calendar = Calendar.getInstance();

            int year = calendar.get(Calendar.YEAR);
            int month = calendar.get(Calendar.MONTH);
            int day = calendar.get(Calendar.DAY_OF_MONTH);

            DatePickerDialog datePickerDialog = new DatePickerDialog(
                    context: MainActivity.this,
                    ( DatePicker view1, int year1, int month1, int dayOfMonth) -> {
                        txtResult.setText("Selected Date: " + dayOfMonth + "/" + (month1+1) + "/" + year1);
                    },
                    year, month, day
            );

            datePickerDialog.show();
        });
```

```java
        // Time Picker
        btnTime.setOnClickListener( View view -> {

            Calendar calendar = Calendar.getInstance();

            int hour = calendar.get(Calendar.HOUR_OF_DAY);
            int minute = calendar.get(Calendar.MINUTE);

            TimePickerDialog timePickerDialog = new TimePickerDialog(
                    context: MainActivity.this,
                    ( TimePicker view12, int hourOfDay, int minute1) -> {
                        txtResult.setText("Selected Time: " + hourOfDay + ":" + minute1);
                    },
                    hour,
                    minute,
                    is24HourView: true
            );
            timePickerDialog.show();
        });
    }
}
```

Output:

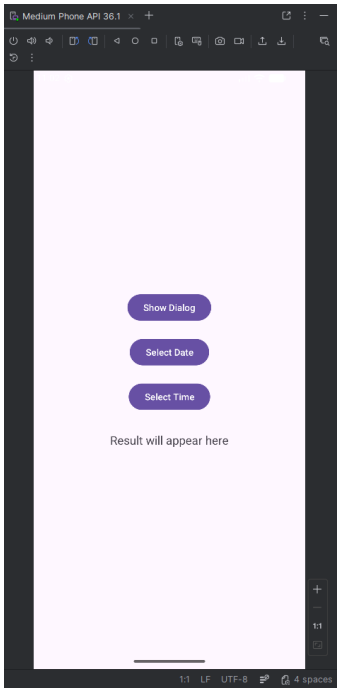
1. Alert Dialog Interaction
   - Displays confirmation message
   - User selects Yes or No
   - Shows Toast response
2. Date Selection
   - Opens calendar interface ● User selects date ● Displays:

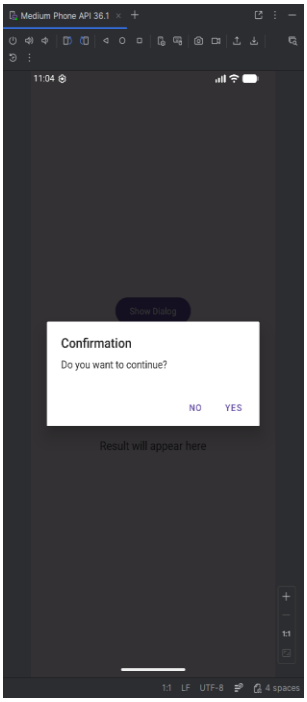
Selected Date: DD/MM/YYYY

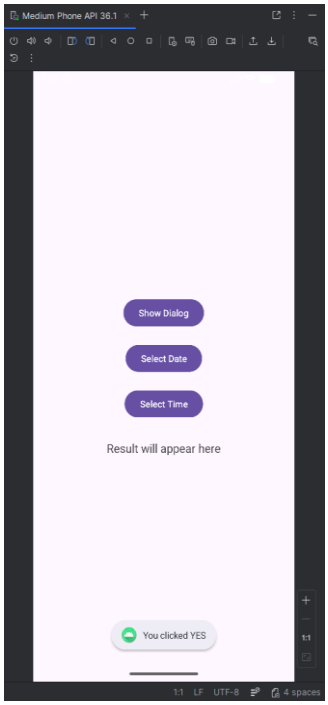
3. Time Selection
   - Opens clock interface ● User selects time ● Displays:

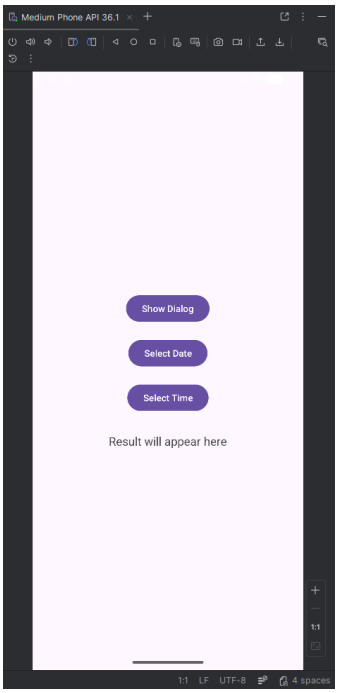
Selected Time: HH:MM

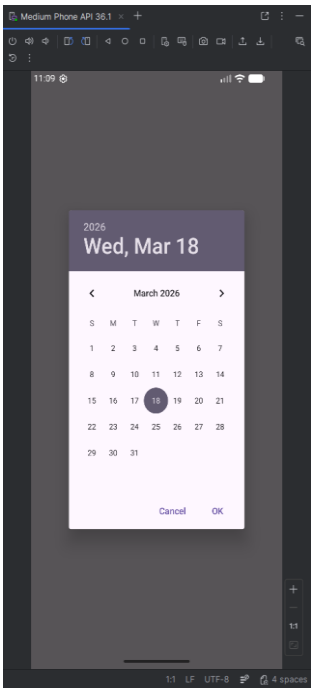
   -
4. Dynamic Output
   - TextView updates based on user actions
5. User-Friendly UI
   - Easy interaction using dialogs and pickers

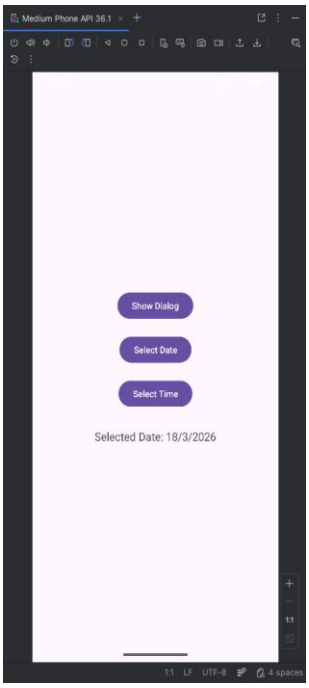

Medium Phone API 36.1
Show Dialog
Select Date
Select Time
Result will appear here

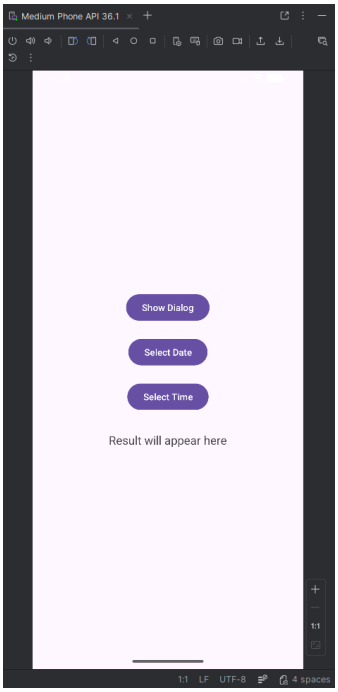

Medium Phone API 36.1
11:04
Confirmation
Do you want to continue?
NO
YES
Result will appear here

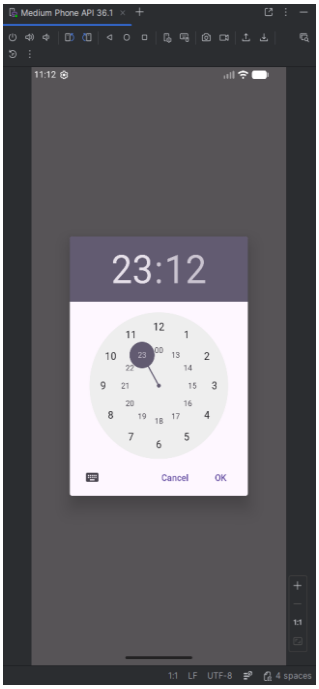

Medium Phone API 36.1
Show Dialog
Select Date
Select Time
Result will appear here
You clicked YES

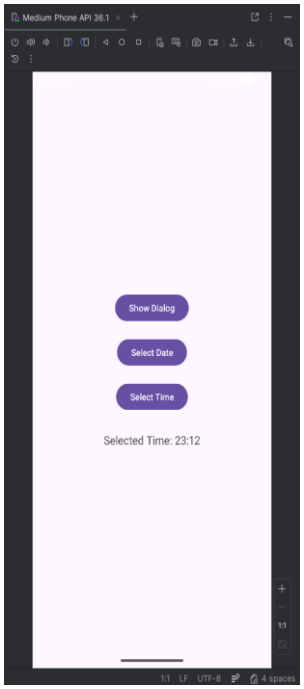

Medium Phone API 36.1
Show Dialog
Select Date
Select Time
Result will appear here


Medium Phone API 36.1
11:09
2026
Wed, Mar 18
March 2026
Cancel
OK


Medium Phone API 36.1
Show Dialog
Select Date
Select Time
Selected Date: 18/3/2026







Conclusion:

This practical demonstrates how Android provides interactive components like dialogs and pickers to improve user experience by allowing easy selection of options, dates, and time.